

SAI PRANAY TUMMALA

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EDUCATION

<i>Robert B. Willumstad School of Business, Adelphi University, Garden City, NY</i> Master of Science in Business Analytics, GPA 3.97/4.0	[Jan. 2024 – May 2025]
<i>International Sri Ramaswamy Memorial Institute of Science and Technology (SRMIST), India</i> Bachelor of Technology in Electronics and Communication Engineering	[June 2014 – May 2018]

PROFESSIONAL EXPERIENCE

Graduate Research and Teaching Assistant <i>Adelphi University, Garden City, NY</i> - Collaborated with the professor for a paper entitled <i>Boosting Techniques in the Machine Learning Algorithms</i>. - Graded assignments, homework, and exams for three distinct courses, ensuring timely and accurate feedback. - Provided academic support by assisting students with course-related inquiries and clarifying complex concepts. - Evaluated and assessed student performance, offering constructive feedback to enhance learning outcomes.	[Jan. 2025 – May 2025]
Metadata Assistant – Library <i>Adelphi University, Garden City, NY</i> - Managing and cataloging the data in the University's Institutional Repository. - Planning and execution of the migration of the publications of various department professors at the university into the Database. - Perform data loads and exports and cleanup using Python and Excel.	[Sept. 2024 – May 2025]
Learning Strategist Coach - Bridges to Adelphi <i>Adelphi University, Garden City, NY</i> - Worked one-on-one with neurodiverse students to improve their executive functioning and time management. - Prepared cohesive weekly schedules and reviewed class syllabi to support students with their assignments. - Generated reports to monitor student progress, tracking key academic metrics to improve performance.	[Aug. 2024 – Jan. 2025]
Quality Specialist <i>Amazon Development Centre, India</i> - Analyzed raw data from ETL pipelines to maintain quality with 96% accuracy, using Excel. - Performed Data Manipulation and Data Analysis on the raw data using MySQL and Excel. - Performed geo-corrections for Address Identification (AID). - Created regular reports to monitor and track process performance, ensuring alignment with business objectives. - Earned performance recognition for contributing to process optimization initiatives that streamlined data reporting, resulting in increased efficiency.	[Sept. 2021 – Dec. 2023]
Transaction Processing Associate <i>Accenture Private Limited, India</i> - Worked for a larger client, reviewing and moderating ads for compliance with policies and Standard Operating Procedures (SOPs). - Processed 40-50 cases per hour with a consistent accuracy rate of 97% to 98%. - Contributed innovative ideas incorporated into SOPs, enhancing efficiency and ad compliance. - Recognized with various awards for exceptional performance and efficiency.	[April 2018 – Jan. 2020]

ACADEMIC PROJECTS

No Show to Appointment - Developed an XGBoost model with LASSO feature selection to predict healthcare appointment no-shows. - Tuned classification threshold (0.5 → 0.6), improving specificity from 75% to 82% and reducing false positives. - Applied class weighting and resampling techniques to address class imbalance, improving model reliability on both show-up and no-show classes. - Achieved 93% precision, 68% sensitivity, and 84% AUC, balancing show-up and no-show predictions. - Used LASSO to identify and incorporate key features (e.g., Age, Waiting Time), boosting model performance. - Reduced unnecessary interventions by accurately predicting no-shows, leading to better resource use.	[March 2025 – May 2025]
Loan Default Prediction Challenge - Developed and tuned multiple machine learning models to predict loan defaults using a dataset of 50,000+ records with demographic and financial features. - Utilized advanced models such as XGBoost, Boosting etc., which helped in achieving finer accuracy using XGBoost (80%). - Conducted extensive hyperparameter optimization and cross-validation (GridSearchCV) to improve model performance. - Evaluated models based on key metrics such as: Accuracy, ROC-AUC, Precision, Recall, F1-Score and Confusion Matrix. - Delivered actionable insights for financial risk management based on model prediction.	[Oct. 2024 – Dec. 2024]
Predicting Potential Prospect Customers – Netrality - Identified high-potential prospects using Clustering Methods (K-Means, Birch, Spectral Clustering), successfully predicting 20+ potential company IDs for targeting from a large prospect customer dataset. - Engineered 15+ new features to enhance model performance, improving the accuracy of prospect identification. - Also, implemented and fine-tuned multiple regression models, optimizing them through 100 iterations to predict key customers. - Conducted comprehensive Exploratory Data Analysis (EDA) on the customers data. - Created actionable data visualizations, highlighting revenue trends by industry and state, directly influencing business decisions and targeted marketing efforts. - Evaluated and selected features for clustering and regression, leading to the prediction of over 30 potential customer IDs, significantly enhancing the prospect targeting process.	[April 2024 – May 2024]

SKILLS

Programming and Data Analysis :-	Python, SQL, Machine Learning Algorithms (Regression, Classification, Clustering, Boosting, Bagging), Data Analysis, Data Cleaning, Data Transformation, Exploratory Data Analysis (EDA), Cross-Industry Standard Process for Data Mining.
Database and Visualization Tools :-	MySQL (Joins, Data Extraction, Querying, Aggregations), Tableau (Storytelling, Dashboards), Quicksight, Excel (Pivot Tables, VLOOKUP), Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn.
Additional Skills :-	Microsoft Office Suit (Outlook , Access, PowerPoint, Visio), JIRA.
Languages :-	English (Fluent), Telugu and Hindi (Native).